

Activity 4 Pre-Lab — Analysis of OTC Pain Relievers (TLC)

[Your name] & [partner]

[MM/DD/YY]

Organic Lab · Introduction to Thin-Layer Chromatography

■ **Black** = write this in your notebook ■ **Blue** = context, don't copy it Copy the header onto every page. Draw anything in a dashed box yourself.

Purpose

Use **thin-layer chromatography (TLC)** to separate and identify the medicinally-active compounds — **acetaminophen**, **aspirin (acetylsalicylic acid)**, and **caffeine** — in an over-the-counter pain reliever (Excedrin). Compare the R_f values of the tablet's spots against authentic standards, and see how the **mobile-phase polarity** changes the separation.

🖋️ **DRAW** — structures of the three analytes

input chemical structure of: **acetaminophen** · **aspirin (acetylsalicylic acid)** · **caffeine** (you'll refer to these when you assign each TLC spot)

Chemical Reagents & Safety Table

CONTEXT ▶ Directions require a table for these 5: ethyl acetate, hexanes, acetaminophen, acetylsalicylic acid, caffeine. Values below are typical Sigma-Aldrich SDS — cross-check the SDS tile on Labflow.

Chemical	Role in lab	MW	Amount used	Physical properties	Hazards (GHS)
Ethyl acetate	Mobile-phase component (polar)	88.11	~20 mL (in mobile phases)	Colorless liquid; bp 77 °C; d 0.90 g/mL; flash pt -4 °C	H225 highly flammable · H319 serious eye irritation · H336 drowsiness/dizziness
Hexanes	Mobile-phase component (non-polar)	86.18	~16 mL (in mobile phases)	Colorless liquid; bp 69 °C; d 0.66 g/mL; flash pt -22 °C	H225 flammable · H304 may be fatal if swallowed/enters airways · H315 skin irritation · H336 drowsiness · H361f repro. tox · H373 organ damage · H411 aquatic tox
Acetaminophen	Analyte / standard	151.16	trace (spotting std)	White crystalline solid; mp 168–172 °C	H302 harmful if swallowed · H315 · H319 · H335
Acetylsalicylic acid (aspirin)	Analyte / standard	180.16	trace (spotting std)	White crystalline solid; mp ~135 °C (dec.)	H302 harmful if swallowed · H315 · H319 · H335
Caffeine	Analyte / standard	194.19	trace (spotting std)	White solid/powder; mp 234–236 °C (sublimes ~178 °C)	H302 harmful if swallowed

FYI ▶ also used (not required in table): methanol (extract solvent — flammable/toxic) and a few drops of glacial acetic acid the instructor adds to each chamber to keep aspirin's acidic spot tight.

GRAB FIRST: mortar & pestle · 25-mL Erlenmeyer · 3× 250-mL beakers + 3 watch-glass covers · 10-mL graduated cylinder · 4 silica TLC plates · capillary spotters · pencil + ruler · forceps · UV lamp.

Procedure Plan (split the page: left = your plan, right = fill in during lab)

Plan (write before lab)

Part 1 — Find the best solvent

- Crush **2 tablets** to powder w/ mortar & pestle → 25-mL Erlenmeyer.
- Add **~5 mL methanol**, swirl 5 min, let the white binder settle. **Clear liquid on top = your extract. Label it "T" (tablet).**
- With a graduated cylinder, make **12 mL each** of three mobile phases in labeled 250-mL beakers; cover each w/ a watch glass:
 - (i) 5:1 hexanes:EtOAc
 - (ii) 1:1 hexanes:EtOAc
 - (iii) 100% EtOAc

Instructor adds 5 drops glacial acetic acid to each; swirl. Watch glass keeps the chamber vapor-saturated.
- Get a 4-lane plate; **lightly** spot "T" on lane 1 of **3** plates (pencil the origin line ~1 cm up first). **Too much sample streaks → unreadable. Re-spot the same dot 2–3× letting it dry between.**
- Stand one plate in each chamber (spots **above** the solvent!), re-cover. Let it run until solvent is ~1 cm from the top.
- Remove, immediately **pencil-mark the solvent front**, air-dry.
- View under the **UV lamp**; pencil-circle every spot. Measure distances, compute R_f , and choose the solvent with the **best-spread** spots.

Part 2 — Identify the components

- New plate, **best solvent**, **4 lanes**: T · acetaminophen · aspirin · caffeine.
- Develop, dry, UV-visualize, circle spots, measure each R_f .
- Match each tablet spot's R_f to a standard → state which actives are present.

✍️ **WRITE the R_f formula**

$R_f = (\text{distance spot moved from origin}) \div (\text{distance solvent front moved from origin})$

TIP ▶ Handle plates by the edges only; skin oils leave spots.

Experiment Notes (in lab)

Tablet brand: _____

Part 1 — R_f per solvent:

5:1 →

1:1 →

0:1 (100% EtOAc) →

Best-resolving solvent: _____

UV notes / any streaking:

Part 2 — R_f match:

acetaminophen std:

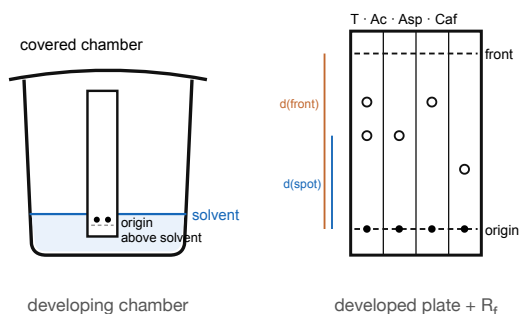
aspirin std:

caffeine std:

tablet spots:

Actives identified in tablet:

Setup — draw these



CONTEXT (don't copy) ▶

Why acid? aspirin is a carboxylic acid; a little acetic acid in the mobile phase keeps it neutral so its spot stays tight.

Polarity: silica = polar stationary phase. More EtOAc (more polar) pushes everything higher (bigger R_f). Pick the solvent that spreads the three spots apart the most.

Clean-up & Conclusions

- Used solvents → **organic (non-halogenated) waste**; TLC plates, capillaries & tablet powder → solid waste; rinse & return glassware; do your assigned bench clean-up task.
- Conclusions** (write ~4 sentences [after lab](#)): which solvent resolved the actives best & why (polarity); the R_f of each standard; which actives your tablet contained by R_f match; one error source (heavy spotting, tilted plate, running the front off).

